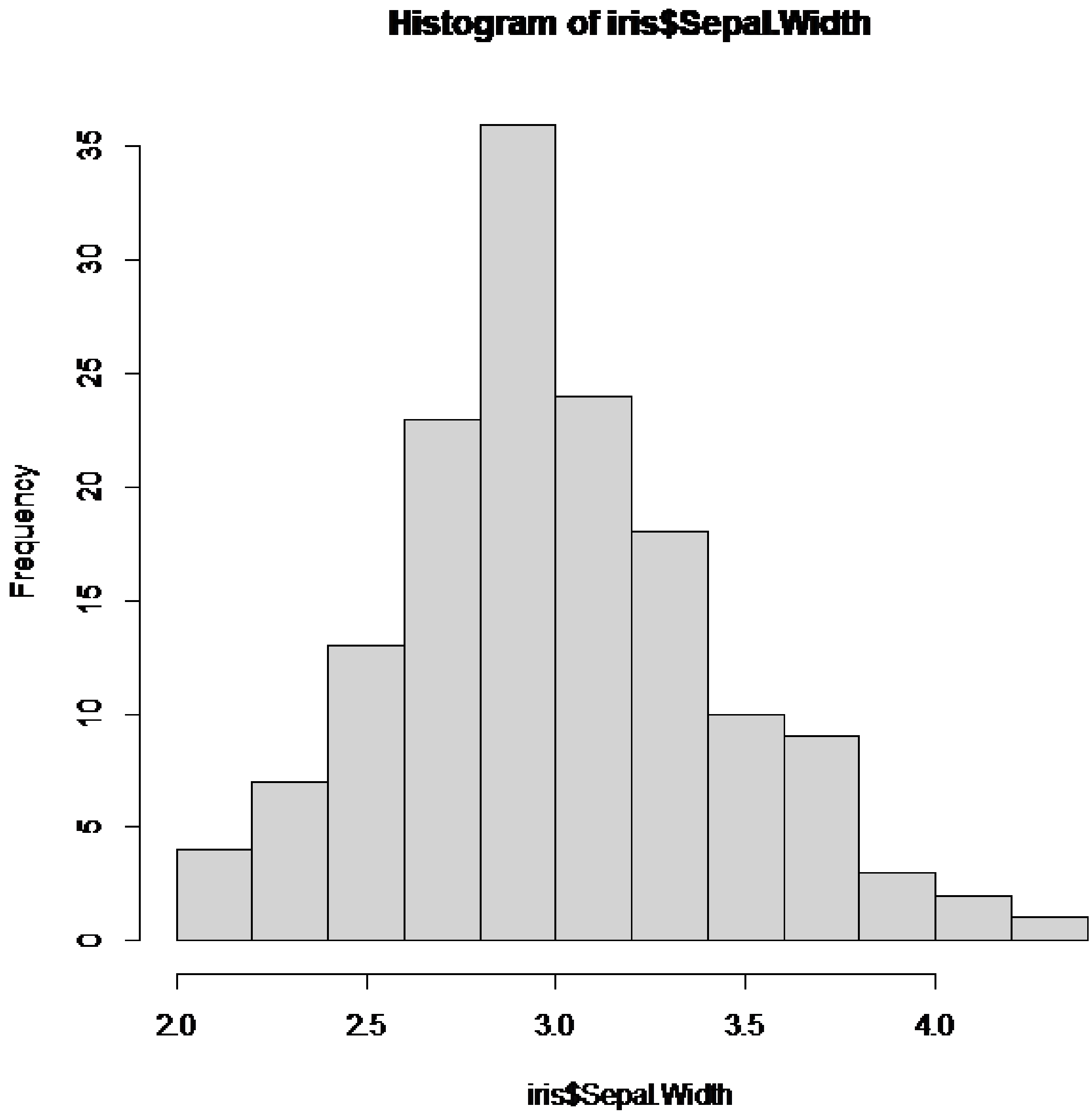
1.

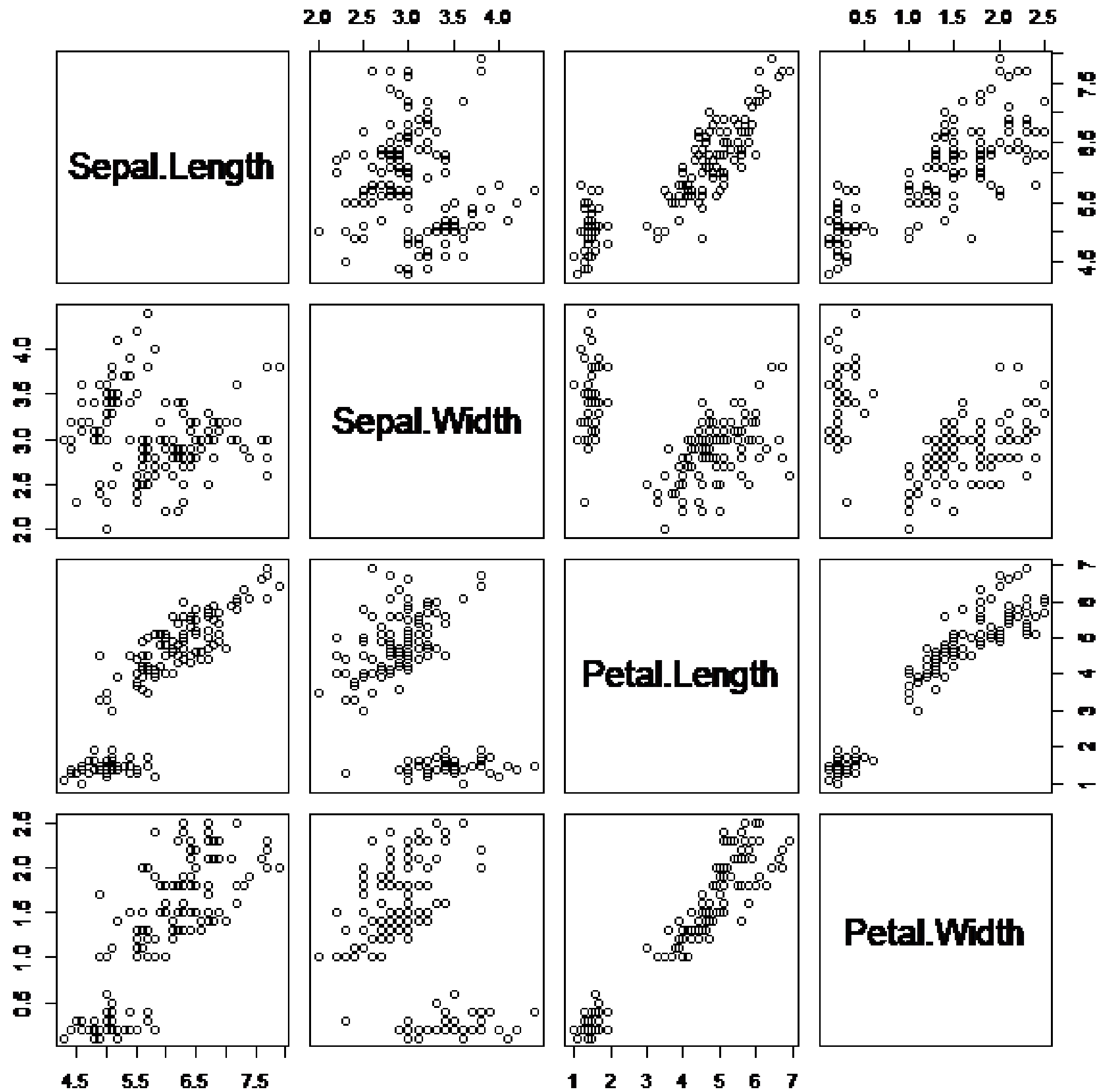
A.



b. The mean is higher because the data skews to the right. Whenever the data skews towards the right the mean is greater than the median.

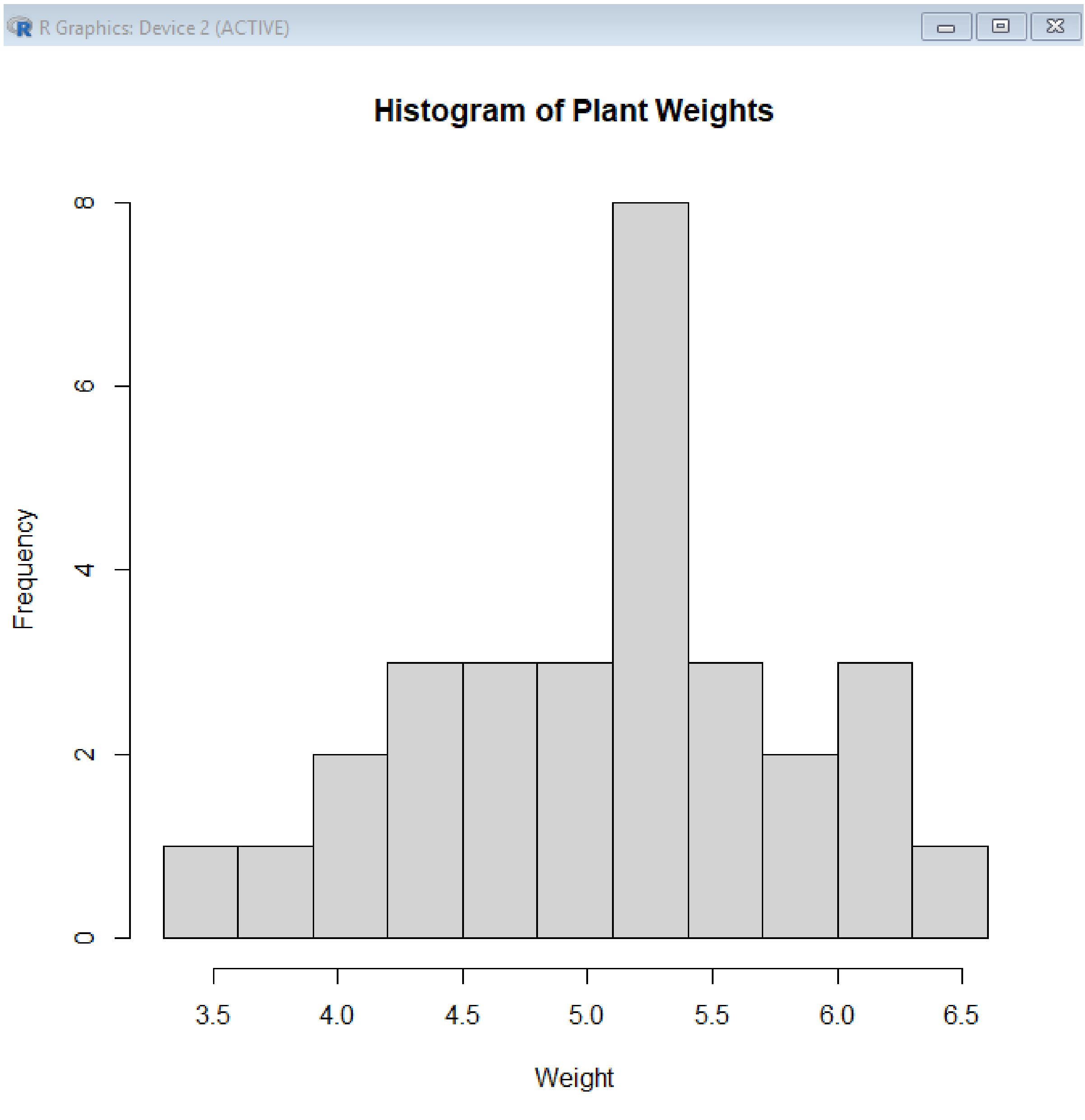
c. The mean of sepal width is 3.057333 and the median is 3.0. Although just slightly larger the mean is in fact greater than the mean.

d. 3.3



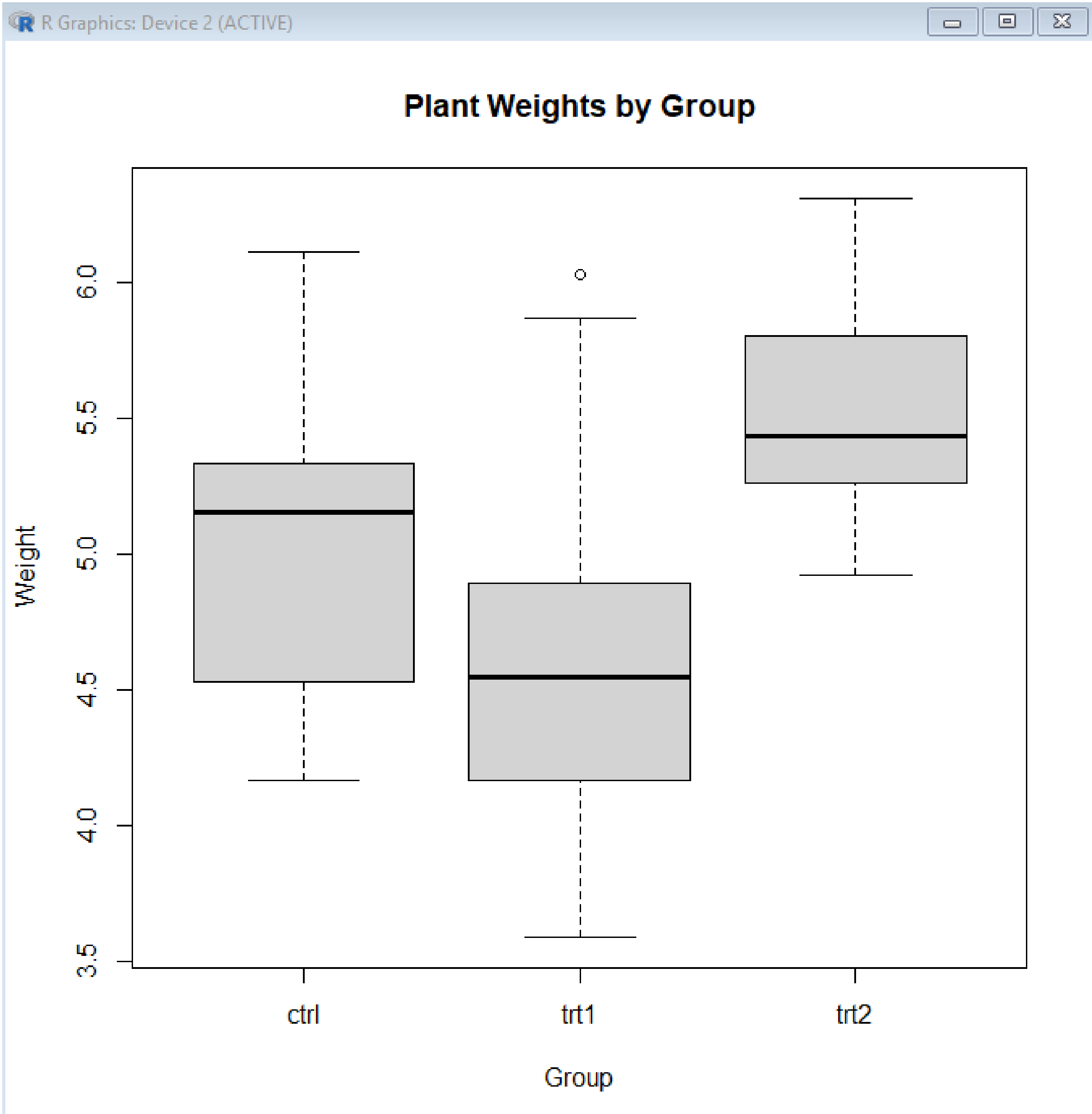
f. Petal.wtdth and Petal.length seem to have a strong relationship. Kind of showing as petal length increase so does petal width. Sepal length and sepal width seem to have a weak relationship.

2.


R Graphics: Device 2 (ACTIVE)
Histogram of Plant Weights
Frequency
0
2
4
6
8
3.5
4.0
4.5
5.0
5.5
6.0
6.5
Weight

A.

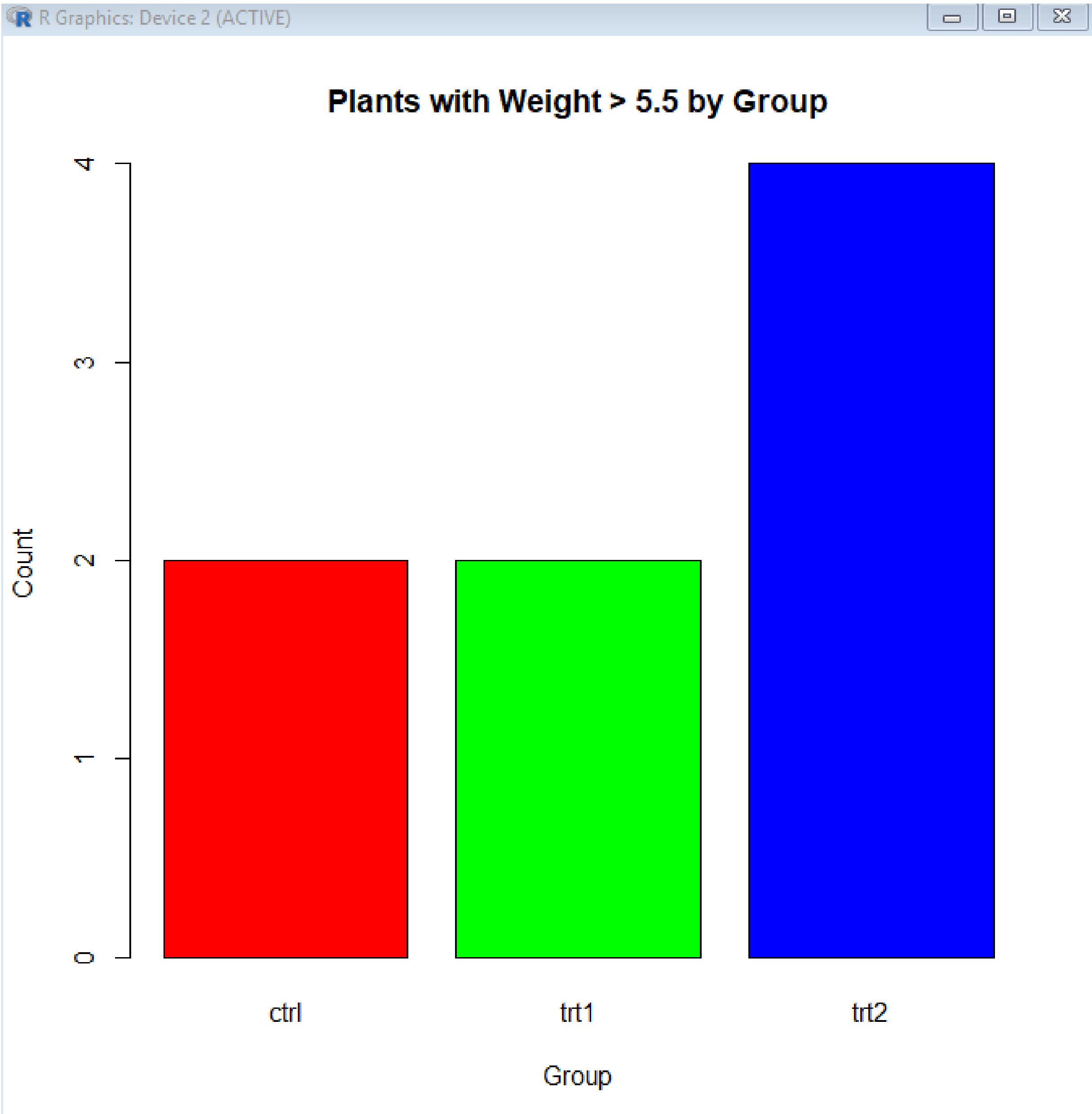
B.



C. From looking at the boxplots aproximately 70-80% of trt1 weights is below trt2 minimum.

D. 80% is the exact amount.

E.



hist(iris$Sepal.Width)

mean(iris$Sepal.Width)

[1] 3.057333

median(iris$Sepal.Width)

[1] 3

quantile(iris$Sepal.Width, 0.73)

73%

3.3

pairs(iris[,1:4])

> pairs(iris[,1:4])

> data(PlantGrowth)

> PlantGrowth

```
   weight group
1    4.17  ctrl
2    5.58  ctrl
3    5.18  ctrl
4    6.11  ctrl
5    4.50  ctrl
6    4.61  ctrl
7    5.17  ctrl
8    4.53  ctrl
9    5.33  ctrl
10   5.14  ctrl
11   4.81  trt1
12   4.17  trt1
13   4.41  trt1
14   3.59  trt1
15   5.87  trt1
16   3.83  trt1
17   6.03  trt1
18   4.89  trt1
```

```
19  4.32  trt1

20  4.69  trt1

21  6.31  trt2

22  5.12  trt2

23  5.54  trt2

24  5.50  trt2

25  5.37  trt2

26  5.29  trt2

27  4.92  trt2

28  6.15  trt2

29  5.80  trt2

30  5.26  trt2

> data(PlantGrowth)

>

> hist(

+   PlantGrowth$weight,

+   breaks = seq(3.3, max(PlantGrowth$weight) + 0.3, by = 0.3),

+   main = "Histogram of Plant Weights",

+   xlab = "Weight",

+   ylab = "Frequency"

+ )

> save.image("C:\\Users\\josea\\Desktop\\R stuff\\Week 3 assignment.R")

> > # Minimum weight in trt2

> min_trt2 <- min(PlantGrowth$weight[PlantGrowth$group == "trt2"])

>

> # Percentage of trt1 weights below that minimum
```

```
> mean(PlantGrowth$weight[PlantGrowth$group == "trt1"] < min_trt2) * 100
[1] 80
> > # Keep only plants with weight > 5.5
> heavy_plants <- subset(PlantGrowth, weight > 5.5)
>
> # Count observations in each group
> group_counts <- table(heavy_plants$group)
>
> # Create a colorful barplot
> barplot(
+   group_counts,
+   col = rainbow(length(group_counts)),
+   main = "Plants with Weight > 5.5 by Group",
+   xlab = "Group",
+   ylab = "Count"
+ )
```